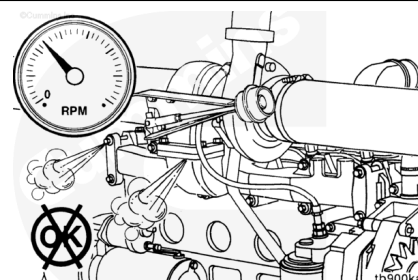


Trainings ▾

General Information

Inspect for exhaust leaks at the exhaust manifold, turbocharger (if equipped), mounting gaskets, or exhaust pipe, muffler or catalyst restrictions. Refer to Procedure 010-024 (/qs3/pubsys2/xml/en/procedures/40/40-010-024-tr.html) and Procedure 010-019 (/qs3/pubsys2/xml/en/procedures/40/40-010-019-tr.html).

For turbocharged engines, leaks or restrictions will cause turbine and impeller to operate at a lower speed and reduce the amount of air being forced into the cylinder. Again, the symptom will be excessive smoke, low manifold pressure, and low power.

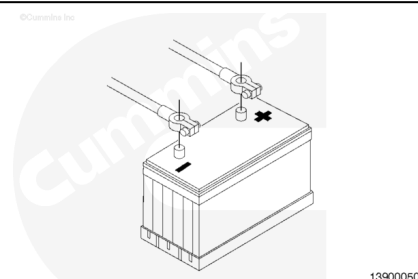


Preparatory Steps

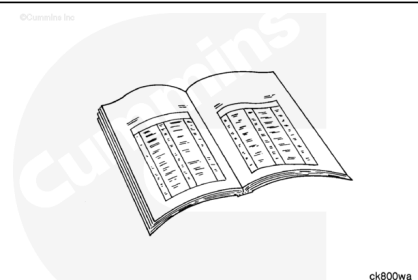
⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Disconnect the batteries.



- If equipped, remove the air crossover. Refer to Procedure 010-019 (/qs3/pubsys2/xml/en/procedures/40/40-010-019-tr.html)
- If equipped, disconnect the charge air cooler piping. Refer to the OEM service manual
- Disconnect the exhaust piping. Refer to the OEM service manual



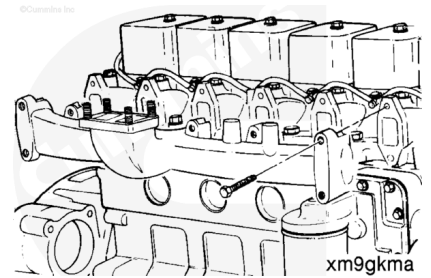
- For turbocharged engines, disconnect the air intake piping. Refer to the OEM service manual
- If equipped, remove the turbocharger. Refer to Procedure 010-033 (/qs3/pubsys2/xml/en/procedures/40/40-010-033-tr.html).

Remove

For 6 cylinder engines, remove the exhaust manifold mounting capscrews and spacers.

Remove the exhaust manifold and gaskets.

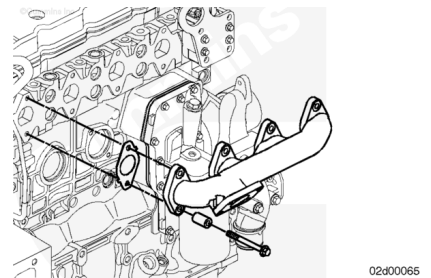
Discard the exhaust manifold gaskets.



For 4 cylinder engines, remove the exhaust manifold mounting capscrews and spacers.

Remove the exhaust manifold and gaskets.

Discard the exhaust manifold gaskets.

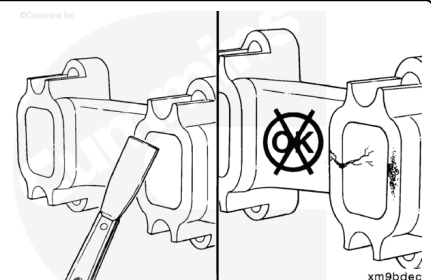


Clean and Inspect for Reuse

Clean the sealing surfaces of the head.

Clean the sealing surfaces of the exhaust manifold.

Use 240-grit inert cloth to remove carbon deposits from the sealing surface.

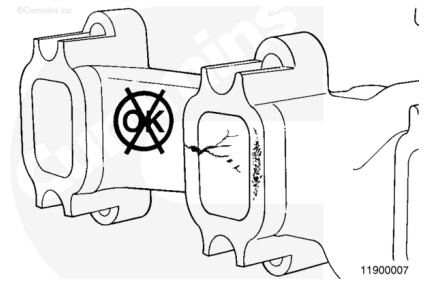


Inspect the parts for cracks, burnout, or damage.

Note : If the exhaust manifold is damaged, for turbocharged engines, check the charge air cooler. A charge air cooler failure can cause progressive damage

to the exhaust manifold. Refer to Procedure 010-027 (/qs3/pubsys2/xml/en/procedures/40/40-010-027-tr.html).

Inspect the gasket surfaces for gouges, scratches, or burnout.



Install

Coat all capscrew threads with an unleaded anti-seize compound.

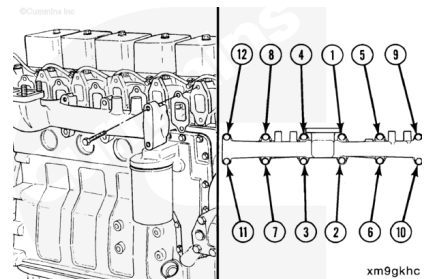


For 6 cylinder engines, install the exhaust manifold, new gaskets, spacers and mounting capscrews.

Follow the tightening sequence shown in the illustration.

Tighten the exhaust manifold mounting capscrews.

Torque Value: 43 n•m [32 ft-lb]

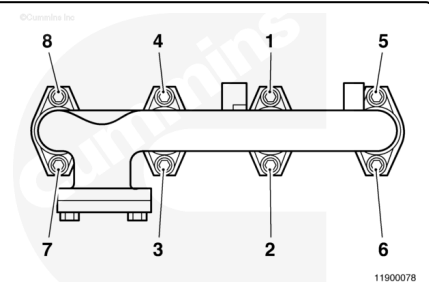


For 4 cylinder engines, install the exhaust manifold, new gaskets, spacers and mounting capscrews.

Follow the tightening sequence shown in the illustration.

Tighten the exhaust manifold mounting capscrews.

Torque Value: 43 n•m [32 ft-lb]

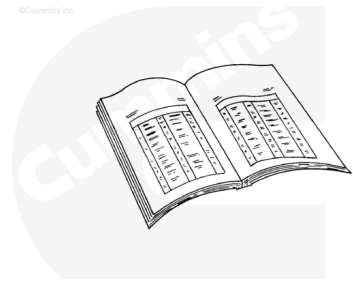


Finishing Steps

- If equipped, install the turbocharger. Refer to Procedure 010-033 (/qs3/pubsys2/xml/en/procedures/40/40-010-

033-tr.html).

- Connect the exhaust piping. Refer to the OEM service manual
- For turbocharged engines, connect the air intake piping. Refer to the OEM service manual
- If equipped, install the air crossover. Refer to Procedure 010-019 (/qs3/pubsys2/xml/en/procedures/40/40-010-019-tr.html)
- If equipped, connect the charge air cooler piping. Refer to the OEM service manual.

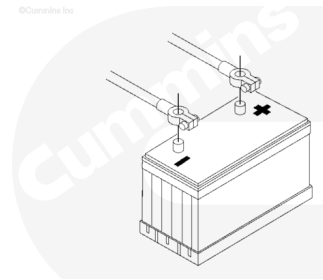


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⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Connect the batteries
- Operate the engine and check for leaks.



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